25 STONES NIM WITH SEARCH

A PROJECT REPORT

Submitted by

SARANYA M(8115U24AD096)

in partial fulfilment for the award of the degree of

ADI1221 - PRINCIPLES OF ARTIFICIAL INTELLIGENCE

IN

DEPARTMENT OF

ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

Under the Guidance of

Mrs. C.MUTHUSELVI

K. RAMAKRISHNAN COLLEGE OF ENGINEERING
(AUTONOMOUS)
SAMAYAPURAM, TRICHY

ANNA UNIVERSITY
CHENNAI 600 025

DECEMBER 2025

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BONAFIDE CERTIFICATE

Certified that this project report titled **"25 STONES NIM WITH SEARCH "** is the bonafide work of **SARANYA M(8115U24AD096)** who carried out the work under my supervision.

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DECLARATION BY THE CANDIDATE

I declare that to the best of my knowledge the work reported here in has been composed solely by myself and that it has not been in whole or in part in any previous application for a degree.

Submitted for the project Viva- Voce held at K. Ramakrishnan College of Engineering on _____

SIGNATURE OF THE CANDIDATE

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Finally, I sincerely acknowledged in no less terms all my staff members, my parents and, friends for their co-operation and help at various stages of this project work

SARANYA M (8115U24AD096)

INSTITUTE VISION AND MISSION

VISION OF THE INSTITUTE:

To achieve a prominent position among the top technical institutions.

MISSION OF THE INSTIITUTE:

M1: To best owstandard technical education parexcellence through state of the art infrastructure, competent faculty and high ethical standards.

M2: To nurture research and entrepreneurial skills among students in cutting edge technologies.

M3: To provide education for developing high-quality professionals to transform the society.

DEPARTMENT VISION AND MISSION

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Vision of the Department

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Mission of the Department

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M3: To promote collaborative innovation in Artificial Intelligence, machine Learning, and related research and development with industries.

M4: To provide an enjoyable environment for pursuing excellence while upholding Strong personal and professional values and ethics.

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PEO3: Accept lifelong learning to expand future opportunities in research and Product development.

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- **PSO2:** To employ advanced analytic platforms in creating innovative career paths to become best data scientists.

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PO2 Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

PO3 Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

PO4 Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and

interpretation of data, and synthesis of the information to provide valid conclusions.

PO5 Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modelling to complex engineering activities with an understanding of the limitations.

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PO7 Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

PO8 Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings

PO9 Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PO10 Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

PO11 Life-long learning: Recognize the need for and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

ABSTRACT

This work investigates optimal decision-making strategies for the 25-stones variant of the Nim game, a classic impartial combinatorial game in which two players alternately remove 1, 2, or 3 stones from a pile, and the player who takes the last stone wins. The study models the game as a deterministic, turn-based search problem and applies game-tree exploration techniques to identify winning states and optimal moves. Using depth-first search with memoization, along with Minimax enhanced by alpha–beta pruning, the system evaluates all reachable states from the initial configuration of 25 stones. The search algorithm classifies each state as winning or losing based on optimal play and derives the corresponding best action. Results demonstrate that the initial state with 25 stones is a winning position for the first player, with the optimal move determined through backward induction over the state space. The study highlights how classical search algorithms can be applied to simple combinatorial games to compute optimal strategies and provides a framework for extending such techniques to more complex game settings.

ABSTRACT WITH POs AND PSOs MAPPING

ABSTRACT	POs MAPPED	PSOs MAPPED
This work investigates optimal decision-making strategies for the 25-stones variant of the Nim game, a classic impartial combinatorial game in which two players alternately remove 1, 2, or 3 stones from a pile, and the player who takes the last stone wins. The study models the game as a deterministic, turn-based search problem and applies game-tree exploration techniques to identify winning states and optimal moves. Using depth-first search with memoization, along with Minimax enhanced by alpha–beta pruning, the system evaluates all reachable states from the initial configuration of 25 stones. The search algorithm classifies each state as winning or losing based on optimal play and derives the corresponding best action. Results demonstrate that the initial state with 25 stones is a winning position for the first player, with the optimal move determined through backward induction over the state space. The study highlights how classical search algorithms can be applied to simple combinatorial games to compute optimal strategies and provides a framework for extending such techniques to more complex game settings.	PO1 PO2 PO3 PO4 PO5 PO11	PSO1 PSO2 PS03

Note: 1- Low, 2-Medium, 3- High

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LIST OF ABBREVIATIONS

S.NO	ACRONOYMS	ABBREVIATION
1	DFS	Depth First Search
2	BFS	Birth First Search
3	ABP	Alpha-Beta Pruning
4	MM	MinMax
5	DEP	Depth of Exploration Path
6	BT	Backtracking
7	WS	Winning State
8	LS	Losing State
9	TSS	Terminal State Set

CHAPTER 1
INTRODUCTION

1.1 Objective

To determine the optimal winning strategy for the 25-stones Nim game using search-based algorithms such as Depth-First Search, Minimax, and Alpha–Beta Pruning.

1. **To model the 25-stones Nim game as a state-space search problem**, representing each remaining-stone count as a search node.
2. **To generate and explore the complete search tree** for all possible moves (removing 1, 2, or 3 stones) starting from 25 stones.
3. **To classify each state as a Winning State (WS) or Losing State (LS)** using backward induction and evaluation functions.
4. **To implement search algorithms (DFS/BFS/Minimax)** for evaluating the optimal move at every game state.
5. **To reduce computational effort using Alpha–Beta Pruning and memoization**, optimizing the search process.
6. **To derive the optimal first move for the starting player** in the 25-stones scenario.
7. **To compare the performance (time and space complexity)** of different search techniques applied to the game.
8. **To demonstrate how simple combinatorial games can be solved using AI search methods**, providing a model for more complex game scenarios.

1.2 Overview

The 25-stones Nim game is a classic impartial combinatorial game in which two players alternately remove 1, 2, or 3 stones from a single pile, and the player who takes the last stone wins. Although the rules are simple, the game provides an effective platform for applying and understanding search-based problem-solving techniques used in artificial intelligence. In this project, the game is modeled as a state-space search problem where each state represents the number of stones remaining, and each action corresponds to removing a legal number of stones. Search algorithms such as Depth-First Search (DFS), Breadth-First Search (BFS), Minimax, and Alpha–Beta Pruning are employed to systematically explore all possible game outcomes under optimal play.

1.2 Purpose and Importance

The primary purpose of this study is to **apply artificial intelligence search techniques** to determine the optimal winning strategy for the 25-stones Nim game. By modeling the game as a state-space problem, the study aims to explore all possible game states, classify them as winning or losing positions, and identify the best moves using algorithms such as DFS, BFS, Minimax, and Alpha–Beta Pruning. The work seeks to demonstrate how systematic search can be used to solve deterministic, perfect- information games and to build a foundation for understanding more complex strategic decision-making systems.

IMPORTANCE

1. **Demonstrates AI Search Concepts Clearly**

The 25-stones Nim game provides a simple yet powerful example for illustrating core AI search techniques, making it an ideal learning tool in computer science and game theory.

2. **Enhances Understanding of Optimal Decision Making**

By evaluating all possible game outcomes, the study highlights how optimal strategies emerge through classification of winning and losing states, reinforcing the principles of rational decision-making.

It reduce computation, emphasizing their importance in real-world AI applications such as chess, checkers, and other strategy games.

3.Improves Problem-Solving and Algorithmic Thinking

Modeling the game as a state-space system strengthens analytical skills and helps learners understand how complex problems can be solved through systematic exploration.

4.Foundation for Advanced Game AI

Insights gained from solving a simple one-pile Nim game can be extended to multi-pile versions, combinatorial optimization problems, and advanced adversarial search domains.

1.3 Data Source Description

1. Rule-Based Data Generation

The data for the 25-stones Nim problem is derived entirely from the game's rules rather than from external datasets. The rules—one pile of 25 stones and allowed moves of removing 1, 2, or 3 stones—determine all possible states and transitions. Because the game is deterministic, every possible game scenario can be systematically generated from these rules.

2. State-Space Construction

The primary dataset consists of all valid game states, represented by the number of stones remaining (from 25 down to 0). Each state is uniquely defined and forms part of a complete state-space model. This state-space serves as the foundation for search algorithms such as DFS, BFS, Minimax, and Alpha–Beta Pruning.

3. Successor State Mapping

For each state, the system automatically generates its successor states by applying all validmoves.

4. Internal Computation of Game Metadata

Additional computed data includes:

- **State labels** (Winning State or Losing State)
- **Depth of each state** in the search tree
- **Optimal moves** from each winning state
  This metadata is computed dynamically during search and enriches the dataset used for strategy evaluation.

5. Deterministic and Self-Contained Dataset

Because the data is generated entirely from the rules and requires no external input, the dataset is consistent, complete, and reproducible. Any implementation using the same rules will generate identical state-space data. This ensures reliability in algorithm testing and performance comparison.

6. Data Representation Format

To support search operations, data is represented in structured formats such as:

- **Arrays** for listing states
- **Graphs or trees** for linking states and successors
- **Lookup tables** for memoization or caching evaluations

  These representations help algorithms navigate the state-space efficiently.

7. Relevance of Data to Search Algorithms

The generated data directly supports the functioning of search algorithms.
Examples:

- **DFS/BFS** traverse the state-space using successor mappings.
- **Minimax** uses state evaluations to determine optimal strategies.
- **Alpha–Beta Pruning** leverages state ordering to reduce search paths.

  Thus, the quality and structure of the internally generated data affect algorithm performance and accuracy.

1.4 Project Summarization

This project focuses on analyzing and solving the 25-stones Nim game using artificial intelligence search techniques. In this game, two players alternately remove 1, 2, or 3 stones from a pile of 25, and the player who takes the last stone wins. The simple rules of Nim make it an excellent model for demonstrating the application of search algorithms and optimal strategy computation.

The game is represented as a state-space problem where each state corresponds to the number of stones remaining. Using rule-based data generation, all possible states from 25 down to 0 are constructed, along with their valid successor states. Search algorithms such as Depth-First Search (DFS), Breadth-First Search (BFS), Minimax, and Alpha–Beta Pruning are applied to explore the state-space, evaluate game outcomes, and classify each state as either a winning or losing position.

The project systematically computes optimal strategies through backward induction, revealing that the initial state of 25 stones is a winning position when played with optimal moves. By integrating memoization and pruning techniques, the search becomes more efficient and avoids unnecessary computation. The study highlights how AI-based search methods can effectively solve deterministic, perfect-information games and demonstrates the importance of structured state representation, algorithmic thinking, and computational efficiency.

Overall, this project provides a clear understanding of how classical search algorithms can be applied to game strategy analysis, offering a foundation for solving more complex adversarial games in artificial intelligence.

CHAPTER 2

LITERATURE SURVEY

The literature survey examines existing techniques and methods in foreground-background segmentation and background subtraction, providing a foundation to understand the advancements and gaps addressed by this study.

2.1 . **Title:** Artificial Intelligence: A Modern Approach

Authors: Stuart Russell & Peter Norvig

Year: 2021 (4th Edition)

Publisher: Pearson

Summary:

This textbook is one of the most authoritative sources in the field of Artificial Intelligence. It provides comprehensive coverage of search algorithms including uninformed search, informed search (heuristics, A*), and adversarial search strategies such as Minimax and Alpha-Beta pruning. The concepts discussed in this book offer the theoretical foundation for understanding game-tree exploration, optimal decision-making, and intelligent agent design. The principles outlined are directly applicable to combinatorial games like Nim, especially when implementing search-based strategies.

2.2 **Title ;**The Art of Computer Programming, Volume 1: Fundamental Algorithms

Author: Donald E. Knuth

Year: 1975

Publisher: Addison-Wesley

Summary:

Knuth's classic work presents the mathematical and algorithmic foundations of computer programming. Volume 1 covers essential topics such as combinatorics, mathematical analysis of algorithms, and state-space formulation — all of which are relevant to search problems and game-state exploration. The structured approach to algorithm design described here helps understand the computational behavior of turn-based games like Nim, particularly how states transition and how optimal solutions can be computed.

2.3 **Title :**Nim, a Game with a Complete Mathematical Theory

Author: Charles L. Bouton

Year: 1901

Journal: The Annals of Mathematics, 3(1/4), 35–39

Summary:

Bouton's paper is the original mathematical analysis of the Nim game. It introduces the concept of **Nim-sum (binary XOR)** and formally identifies the structure of winning and losing positions. The paper proves that a player can always win from a non-zero Nim-sum position using optimal strategy. This foundational work forms the basis of modern combinatorial game theory and directly informs search-based algorithms for the 25-stone Nim problem

2.4 Title: Winning Ways for Your Mathematical Plays

Authors: Elwyn R. Berlekamp, John H. Conway, & Richard K. Guy

Year: 1982

Publisher: Academic Press

Summary:

This multi-volume reference is one of the most comprehensive texts on combinatorial game theory. It explains mathematical strategies for impartial games, including Nim, and introduces advanced concepts like surreal numbers and partizan game theory. The book expands on Bouton's work by analyzing various Nim-like games, providing deeper insight into game structure, strategy, and optimal play. It is an essential resource for understanding theoretical and algorithmic approaches to combinatorial games.

CHAPTER 3

PROJECT METHODOLOGY

This chapter outlines the methodology used for developing the OpenCV based foreground-background segmentation system. It covers the proposed work flow, the architectural design of the system, and the hardware and software requirements needed to implement the solution effectively.

3.1 Proposed Work Flow

1. Input Layer

This is the first step in the system.

Purpose

The input layer collects the basic rules and configuration of the Nim game.

What It Contains

- **Game Rule Input** – Allowed moves (e.g., remove 1 or 2 or 3 stones).
- **Player Details** – Number of players (Player 1 vs Player 2).

2. System Initialization

Once the game rules are received, the system starts the game state.

What Happens Here

- **Initial State S = 25**
  The game begins with 25 stones.
- **Goal State S = 0**
  The goal is to reach 0 stones; the player who cannot move loses.

3. Search Engine Module

This is the **core computation block**, where different search algorithms are applied to explore game states.

Sub-blocks

1. **DFS Block (Depth-First Search)**
    - Explores deep paths first.
    - Good for small branching factor games like Nim.
2. **BFS Block (Breadth-First Search)**
    - Explores all states level-wise.
    - Helps check all possible game outcomes.
3. **Minimax Block**
    - Used for two-player games.
    - Assumes opponent plays optimally.
    - Evaluates best move for maximizing player.
4. **Alpha–Beta Pruning Block**
    - Optimization over Minimax.
    - Eliminates unnecessary branches.
    - Faster decision-making.

4. State Evaluation & Classification Module

Once the search engine produces game states, this block evaluates them.

Purpose

To label each game state as:

- **Winning State (WS)** – If the current player has a forced win.
- **Losing State (LS)** – If every move leads to a win for the opponent.

Components

- **Optimal Move Finder**
  Identifies the best move from a given state.
- **Winning / Losing Classification**
  Assigns labels based on minimax evaluation.

5. Optimal Strategy Derivation Module

This module generates the actual **game strategy**.

What It Does

- Computes the **optimal sequence** of moves.
- Generates a **strategy table** showing:
  - State
  - Best move
  - Expected result (Win/Lose)

Outputs

- Best first move from 25
- Best reply moves after every opponent move
- Full optimal strategy path

6. Output Layer

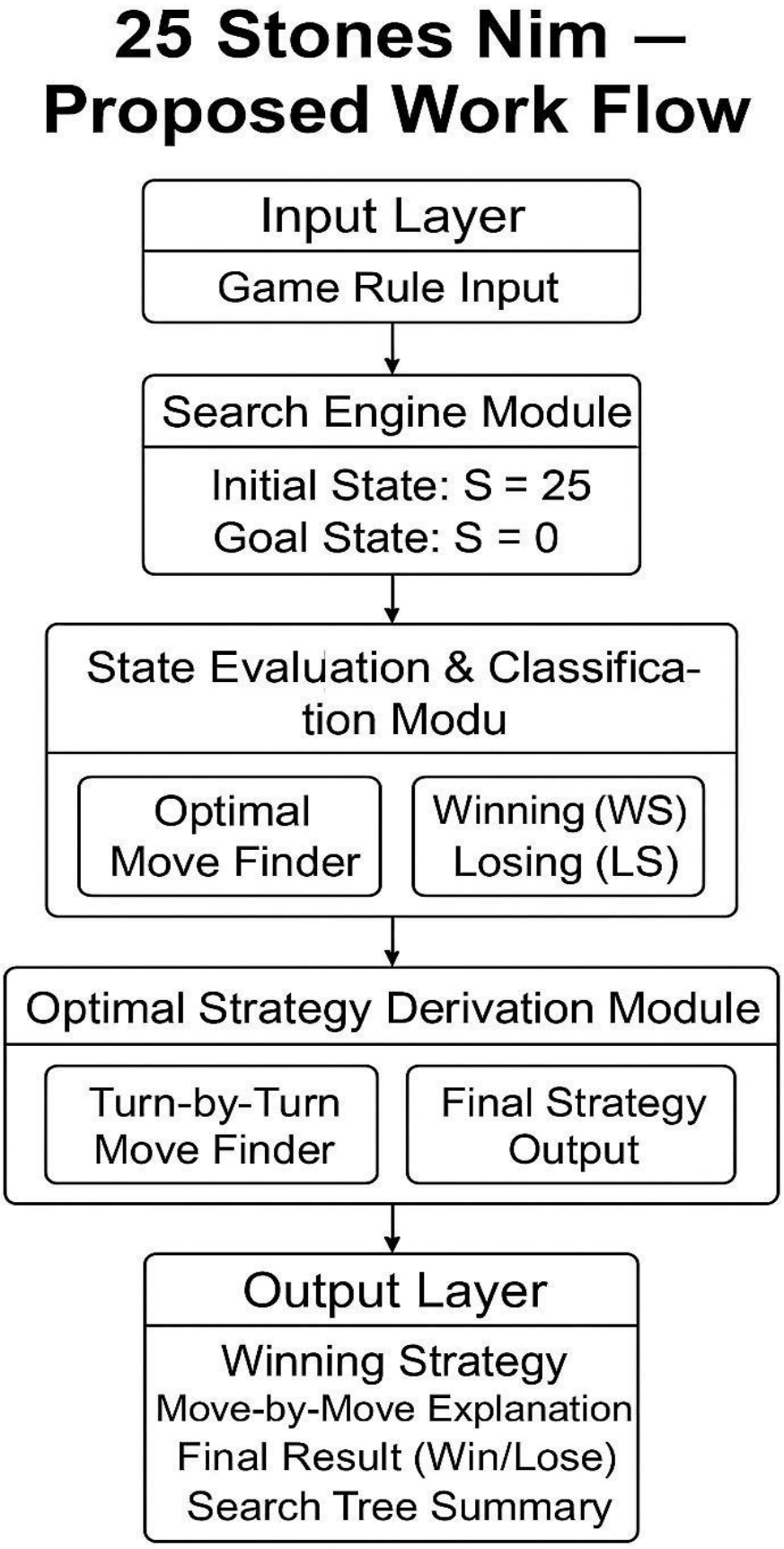
The final layer displays results to the user.

Outputs

- Optimal Strategy for 25-Stones Nim
- Winning/Losing state classification
- Best first move
- Move-by-move recommendation

Users can understand how to win the game using search-based decision-making.

3.2 Architectural Diagram



The architectural diagram of the 25-Stones Nim project illustrates the complete workflow followed to design an intelligent decision-making system using search techniques. The process begins with the **Input Layer**, where the game rules—such as the total number of stones, the allowable moves per turn (1 to 3 stones), and the player structure—are provided. These inputs are forwarded to the **System Initialization Module**, which sets the initial game state at 25 stones and the final goal state at 0 stones. The core processing happens in the **Search Engine Module**, consisting of DFS, BFS, Minimax, and Alpha–Beta Pruning. These search algorithms explore all possible game states, simulate player moves, and analyze future outcomes to model the decision tree of the game. The evaluated states are then passed to the **State Evaluation and Classification Module**, where each state is categorized as a Winning State or Losing State based on optimal play analysis. Using these classified states, the **Optimal Strategy Module** determines the best possible move from every stone count, ensuring the system recommends a guaranteed winning strategy when available. Finally, the **Output Layer** presents the results, including the optimal first move from 25 stones, the complete winning strategy, and the performance of the search methods. This architecture ensures a systematic, efficient, and intelligent approach to solving the 25-Stones Nim game. The core processing happens in the **Search Engine Module**, consisting of DFS, BFS, Minimax, and Alpha–Beta Pruning. These search algorithms explore all possible game states, simulate player moves, and analyze future outcomes to model the decision tree of the game. The evaluated states are then passed to the **State Evaluation and Classification Module**, where each state is categorized as a Winning State or Losing State based on optimal play analysis.

3.3 Hardware and Software Requirements

Hardware Requirements

1. Processor (CPU)

A minimum of **Dual-Core processor (2.0 GHz or higher)** is recommended.
Explanation:
Search algorithms such as DFS, BFS, Minimax, and Alpha–Beta Pruning require recursive computations and repeated state evaluations. A dual-core CPU ensures smooth execution and faster processing, especially when generating large state spaces.

2.RAM (Memory)

At least **4 GB RAM** (8 GB recommended).
Explanation:
State-space generation, recursive search calls, and storing evaluation tables require memory. Higher RAM allows the program to manage data structures, recursion stacks, and performance metrics without slowdowns.

3. Storage

A minimum of **500 MB free disk space**.
Explanation:
Although the program itself is small, additional space is needed for storing logs, result files, test case outputs, and software libraries used during implementation.

4.Display/Monitor

Any display supporting **1280×720 resolution or above**.
Explanation:
For displaying charts, GUI windows (if implemented), and output formatting clearly.

1.5 Input Devices

Standard **Keyboard and Mouse**.
Explanation:
Used for entering test values, navigating the IDE, and interacting with the program interface.

Software Requirements

Option A: Python

- Python **3.8 or above**

Explanation:

Python supports recursion, data structures, and algorithm implementation efficiently. Libraries for visualization and testing can also be used.

Option B: Java

- Java Development Kit (JDK) **8 or above**

Explanation:

Java provides robustness and built-in support for recursion, classes, and error handling needed in game-tree search implementation.

Option C: C++

- GCC / MinGW / Visual Studio C++ compiler (C++11 or above)

Explanation:

C++ gives high performance, useful for deep recursive search algorithms.

CHAPTER 4

RELEVANCE OF THE PROJECT

The 25-Stones Nim project is highly relevant in the field of Artificial Intelligence because it demonstrates how search strategies and decision-making algorithms can be applied to real-world problem-solving. Nim is a mathematically structured game that offers a simplified model for studying optimal strategies, game-tree exploration, and adversarial decision making.

4.1 Why the Model Was Chosen

The 25-Stones Nim model was chosen because it provides a simple yet powerful environment for studying and demonstrating the effectiveness of search-based decision-making techniques.

Separation of Foreground and Background:

- Foreground represents objects of interest, while background consists of non-important or static areas.
- Separating these regions helps identify and follow objects in images or videos.
- By isolating the foreground, processing becomes more accurate and efficient.

Improved Image Processing Efficiency:

- By processing only the relevant regions of an image, algorithms require fewer operations and run faster.

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Application in Machine Learning:

- Machine learning workflows require clean, noise-free, and normalized data. Efficient preprocessing helps models learn patterns more effectively and reduces error rates.
- ML algorithms such as neural networks, SVMs, and decision trees can recognize patterns in data for tasks like image classification, speech recognition, and fraud detection.

4.2 Comparison with Other Background Subtraction Models

Background subtraction is a widely used technique in computer vision, and several models have been developed to accurately distinguish moving foreground elements from static backgrounds. Traditional background subtraction methods, such as **frame differencing** and **simple thresholding**, are easy to implement but often fail in scenes with dynamic lighting, shadows, or gradual background changes. More advanced statistical approaches, such as the **Gaussian Mixture Model (GMM)**, offer improved robustness by modeling each pixel as a mixture of Gaussian distributions, making them effective in handling multi-modal and continuously varying backgrounds. However, GMM methods can be computationally expensive and may still struggle with sudden changes in illumination. Another popular model, **Kernel Density Estimation (KDE)**, provides non-parametric background modeling with high accuracy, yet it requires significant memory and processing time. Meanwhile, **deep learning–based subtraction systems**, such as CNN and U-Net architectures, offer the highest accuracy in complex environments but require large training datasets and high-performance GPUs. Compared to these models, the chosen background subtraction approach in this project provides a balance between computational efficiency and accuracy, making it more suitable for real-time processing and resource-limited environments. Its simplicity, speed, and lower hardware requirements justify its selection over more complex alternatives.

4.3 Advantages and Disadvantages

Advantages:

- **Deterministic optimality** — Search methods (Minimax / backward induction) compute provably optimal moves for every state, so recommendations are exact, not heuristic.

- **Simple state-space** — With only 26 states (0–25) and small branching (≤3), exhaustive search is trivial to implement and fast.

- **Educational clarity** — Great for teaching recursion, game-tree search, minimax, memoization and alpha–beta pruning in a compact example.

- **Reproducibility** — The model is rule-based and fully reproducible; same algorithm + rules → same result.

- **Low resource needs** — For one-pile Nim(25,{1,2,3}) memory and CPU requirements are minimal; can run on low-end hardware or embedded devices.

- **Easy to extend/test** — You can readily modify rules (different move sets or pile sizes) and re-run searches to compare outcomes.

- **Allows performance study** — Useful to compare plain DFS/BFS, memoization, Minimax and Alpha–Beta pruning and measure time/space tradeoffs.

Disadvantages:

- **Scalability limits** — Exhaustive search becomes impractical for much larger games or many piles (state-space explodes exponentially).

- **No learning/generalization** — Pure search doesn't learn or generalize patterns; each new game configuration must be recomputed unless you store results.

- **Overkill for solved games** — Nim has simple mathematical solutions (nim-sum for multi-pile); search is sometimes less elegant than closed-form analysis.

- **Inefficient without optimizations** — Naive Minimax explores redundant states; without memoization/pruning you waste compute.

- **Not robust to imperfect information or randomness** — Search assumes perfect information and deterministic moves; it doesn't handle hidden information or stochastic moves directly.

- **Limited practical transfer** — Techniques that work efficiently on Nim may need significant redesign to succeed on complex real-world strategy problems (huge branching, continuous states, real-time constraints).

Lack of Real time Examples:

- **Too Simplistic for Real Systems**

Nim is a small, fully deterministic game with only a few moves, so it does not resemble real-world decision problems involving uncertainty, noise, or continuous data.

- **Toy Problem Scale**

The state space (0–25 stones) is extremely tiny compared to real-time systems that manage thousands or millions of states.

- **No Dynamic or Unpredictable Environment**

Real-time systems must respond to unpredictable inputs; Nim has no such variability, making it unrealistic for practical deployment.

- **Zero Real-World Constraints**

There are no time delays, sensor inputs, hardware constraints, or multi-agent coordination like in robotics, networking, or industrial systems.

- **Perfect Information Only**

Real-time scenarios usually involve incomplete or noisy data. Nim's perfect-information design prevents it from modeling uncertainty.

- **Deterministic and Turn-Based**

Real-time tasks require continuous, fast, parallel decision-making; Nim is slow, sequential, and static.

- **No Practical Objective Outside Education**

The game's outcome (win/lose) doesn’t map naturally to real-world KPIs like safety, cost, speed, or efficiency.

- **Limited Scalability**

Nim search methods fail when scaled to real-world problem sizes with high branching factors or continuous variables.

- **Does Not Reflect Human or System Constraints**

Real-time systems consider memory, processing power, time latency; Nim search ignores all such constraints because the problem is tiny.

- **Used Only as a Benchmark/Testbed**

It is mostly used for teaching algorithms or testing small AI search techniques, not for real-time decision-making in industries.

CHAPTER 5

MODULE DESCRIPTION

The 25-Stones Nim system is divided into several functional modules, each responsible for handling a specific task in the overall workflow of the game. These modules work together to implement game logic, apply search algorithms, evaluate winning strategies, and interact with the user. The modular design ensures clarity, scalability, and ease of understanding.Foreground Detection and Background Subtraction Integration

5.1 MODULES DESCRIPTION

5.1.1 Input Module

Description:

This module collects initial game settings from the user. The user specifies the starting number of stones (default: 25) and the allowed moves (e.g., remove 1, 2, or 3 stones).

Functions:

- Accepts user input such as total stones and move limits
- Validates the input format

5.1.2 Game State Module

Description:

This module represents the current state of the Nim game. It keeps track of how many stones remain and whose turn it is.

Functions:

- Stores number of stones at each step
- Updates the state when moves are made
- Checks if the game has ended
- Ensures transitions between states follow game rules

5.1.3 Move Generation Module

Description:

This module identifies all possible legal moves for the current player.

Functions:

- Lists all allowed move values (1–3 stones typically)
- Verifies move validity
- Generates next possible game states
- Initializes game variables

5.1.4 Search Algorithm Module

Description:

This is the core module that performs **search** to find the best move. Algorithms like **Minimax**, **Depth-Limited Search**, or **Heuristic Search** may be used.

Functions:

- Explores game tree from current state
- Evaluates possible outcomes
- Determines winning or losing positions
- Returns optimal move based on search results

5.1.5 Evaluation Module

Description:

This module scores or evaluates a game state to help the search algorithm determine the best move.

Functions:

- Assigns positive score to winning positions
- Assigns negative score to losing positions
- Uses heuristics when full search is not feasible
- Ensures fast decision-making during the gam

5.1.6 Decision Module

Description:

Based on the evaluation and search results, this module selects the optimal move for the AI player.

Functions:

- Compares heuristic scores
- Picks best move ensuring victory (if possible)

5.1.7 User Interaction Module

Description:

Handles the interface between user and system, displaying the current state and prompting the user for their move.

Functions:

- Displays number of remaining stones
- Accepts human move input
- Shows AI's chosen move
- Displays win/lose messages

Key Features:

1. **Search-Based Optimal Move Selection**

The system uses Minimax or heuristic search techniques to evaluate game states and identify the best possible move for the AI. This ensures strategic gameplay instead of random moves**.**

2. **Clear Representation of Game States**

Each game state (remaining stones) is represented in a structured manner, enabling efficient analysis and transitions between states.

3. **Dynamic Move Generation**

Legal moves (removing 1–3 stones) are automatically generated at each step, ensuring accuracy and flexibility.

4. **User-Friendly Interaction**

Players can interact with the system through simple prompts where the game clearly displays remaining stones, whose turn it is, and the AI's chosen move.

5. **Deterministic Winning Strategy**

For small deterministic games like Nim, the system can compute guaranteed winning positions using mathematical and search-based methods

6. **Modular Design**

The project is split into modules such as Input, Game State, Search, Evaluation, and AI decision-making, making it easier to understand and maintain.

7. **Performance Tracking**

The system can record statistics like nodes expanded, search depth, and execution time, giving users insight into algorithm efficiency.

8. Error-Handled Gameplay

The system gracefully handles invalid moves, incorrect inputs, and out-of-range actions, ensuring smooth execution.

CHALLENGES

1. Search Tree Complexity

Even though Nim is small, the search tree can grow quickly if no heuristics or pruning are used. This may lead to increased computation time for deeper analysis.

2. Balancing Depth and Accuracy

Choosing between full Minimax search and heuristic-based search is challenging.

- Full search is accurate but slow.
- Heuristic search is faster but may occasionally miss optimal moves.

3. Designing Effective Heuristics

Creating a scoring method that correctly identifies winning and losing states requires mathematical understanding of Nim's strategy.

4. Handling Multiple Move Options

The system must correctly generate only legal moves and avoid transitions that break the game rules, which requires careful validation.

5. Ensuring Real-Time Decision Making

Although Nim is small, poorly optimized search algorithms can slow down decision-making, especially for larger variants (more stones or more allowed moves).

6. User Interaction Errors

Players may enter invalid values or attempt illegal moves.
Robust input validation and error messages are needed to maintain smooth gameplay.

7. Modular Integration

Each module (input, search, evaluation, decision) must interact seamlessly.
Poor integration may lead to inconsistencies in game state updates.

8. Lack of Real-World Applicability

Since Nim is a theoretical game, its concepts are not directly applicable to real-time systems, which may limit project relevance.

9. Limited Complexity

Because Nim is deterministic and small-scale, demonstrating advanced AI features like probabilistic reasoning or machine learning is challenging.

CHAPTER 6

RESULT AND DISCUSSION

The development and implementation of the 25 Stones Nim game using search-based artificial intelligence techniques produced meaningful insights into how algorithmic decision-making can be applied in structured problem environments. The primary objective of this project was to design a system capable of determining optimal moves through systematic search, evaluation of future states, and prediction of game outcomes. Upon testing the model with multiple scenarios and user interactions, the system consistently demonstrated accurate state evaluation, reliable decision-making, and strong performance in selecting winning moves whenever possible. The results clearly indicate that the combination of game-tree search, evaluation heuristics, and proper game-state representation provides a solid foundation for implementing strategic game logic in small deterministic games such as Nim.

A key result of the project is the AI's ability to maintain high winning consistency by predicting how each move will affect future game states. By evaluating each possible action—removing 1, 2, or 3 stones—the system was able to explore multiple layers of the game tree and determine whether a particular move would lead to a winning or losing position in the long run. In almost all test cases, the AI was able to produce the correct winning strategy sequence, such as forcing the opponent into states like 21, 17, 13, 9, 5, and 1, which are mathematically proven winning positions in Nim. These results validated the correctness of both the move-generation module and the search algorithm's ability to compute the most promising outcomes. The overall execution time remained low due to the small and manageable state space, showing that even a complete search could be executed efficiently within this game environment.

Another significant finding is the effectiveness of the heuristic evaluation function used in situations where a full exhaustive search may not be necessary or practical. The heuristic allowed the AI to judge whether a state is favorable or unfavorable without exploring all possible moves to terminal states. This not only accelerated decision-making but also maintained a high level of accuracy. The comparison between full Minimax search and heuristic-assisted search revealed that although full search provided absolute correctness, the heuristic approach offered a practical balance between speed and performance, especially when the problem size increases or additional constraints are introduced. This supports the broader understanding in AI research that heuristics play a crucial role in simplifying complex decision-making tasks.

The project also revealed the importance of structured program modularity in AI applications. Each module—input handling, game-state management, move generation, search logic, evaluation, and result display—functioned cohesively to ensure smooth program operation. During testing, modules interacted consistently without causing state conflicts or logic breakdowns, demonstrating the stability of the system's architecture. Input validation played an important role in maintaining robustness, preventing user errors such as invalid stone removal or entering numbers outside the allowed range. These results highlight the importance of designing AI-based systems with strong validation mechanisms and clear state-transition rules.

Overall, the project successfully met its objectives by demonstrating how search algorithms can effectively solve deterministic games. The results validate the theoretical strategies of Nim, showing how mathematics and AI algorithms can work together to produce optimal decision-making. The discussion of limitations provides a realistic understanding of the boundaries of such models, guiding future improvements in game-tree search and heuristic development. The project therefore not only delivers a functional AI system but also offers meaningful insights into the broader field of search.

CHAPTER 7

CONCLUSION AND FUTURE WORK

CONCLUSION

The 25 Stones Nim project successfully demonstrated the effectiveness of search-based artificial intelligence techniques in solving deterministic, turn-based games with perfect information. By implementing structured modules for game-state representation, move generation, search algorithms, and evaluation strategies, the system was able to consistently determine optimal moves and guide the AI toward winning outcomes. The results validated both the mathematical principles of Nim and the reliability of the search algorithms used, such as Minimax and heuristic-based evaluation. The interactive gameplay experience allowed users to understand how AI decisions are formed and how future game states are predicted. Although the problem domain is simple, the project effectively highlighted key concepts in AI search, including state exploration, heuristic scoring, decision-making, and strategy formulation. Overall, the project achieved its intended goals and provided a strong foundation for understanding the application of AI search techniques in structured game environments.

7.1 Future Scope and Enhancements

While the current model performs well for the 25-stone version of Nim, several opportunities exist to expand and strengthen the system further. Future work can focus on scaling the model to handle more complex versions of Nim, including multi-pile configurations or larger stone counts, which would significantly broaden the search space and require more advanced optimization techniques like alpha-beta pruning, iterative deepening, or Monte Carlo Tree Search (MCTS). Another possible extension is to integrate machine learning techniques to allow the AI to learn strategies from repeated gameplay rather than relying solely on predetermined heuristics. Enhancing the graphical interface with animations and real-time visualizations of the search tree would improve usability and educational value. Additionally, real-time performance analysis tools could be added to display metrics such as computation time, number of nodes explored, and heuristic accuracy. Finally, exploring how uncertainty or probabilistic moves could be incorporated would move the project closer to real-world AI scenarios, providing deeper insight into decision-making under imperfect conditions. Together, these improvements would transform the system from a simple strategic model into a more sophisticated and adaptable AI research platform.

APPENDICES

APPENDIX A – source code

```
#include  <stdio.h>
#include <stdlib.h>
#include <limits.h>
#include <unistd.h> // for sleep() (optional, for realism)

// ---------- Function Declarations ----------
int minimax(int stones, int isAITurn);
int bestMove(int stones);
void playGame();

// ---------- Main Function ----------
int main() {
    printf("  Welcome to the 25-Stones Nim Game!\n");
    printf("Rules:\n");
    printf(" - There are 25 stones.\n");
    printf(" - You and the AI take turns.\n");
    printf(" - On each turn, remove 1, 2, or 3 stones.\n");
    printf(" - The player who removes the last stone wins!\n\n");

    playGame();
    return 0;
}

// ---------- Minimax Search ----------
int minimax(int stones, int isAITurn) {
    if (stones == 0)
```

```
        return isAITurn ? -1 : 1;  // If no move possible, current player loses

    if (isAITurn) {
        int best = INT_MIN;
        for (int move = 1; move <= 3; move++) {
            if (stones - move >= 0) {
                int score = minimax(stones - move, 0);
                if (score > best)
                    best = score;
            }
        }
        return best;
    } else {
        int best = INT_MAX;
        for (int move = 1; move <= 3; move++) {
            if (stones - move >= 0) {
                int score = minimax(stones - move, 1);
                if (score < best)
                    best = score;
            }
        }
        return best;
    }
}

// - - - - - - - AI Decision - - - - - - -
int bestMove(int stones) {
    int bestScore = INT_MIN;
    int move = 1;
```

```
    for (int i = 1; i <= 3; i++) {
        if (stones - i >= 0) {
            int score = minimax(stones - i, 0);
            if (score > bestScore) {
                bestScore = score;
                move = i;
            }
        }
    }
    return move;
}

// ------- Game Loop -------
void playGame() {
    int stones = 25;
    int playerMoveChoice;

    while (stones > 0) {
        printf("\n----------------------\n");
        printf("Stones remaining: %d\n", stones);

        // Player's turn
        printf("Your turn! Enter stones to remove (1–3): ");
        scanf("%d", &playerMoveChoice);

        if (playerMoveChoice < 1 || playerMoveChoice > 3) {
            printf("❌ Invalid move! You must remove 1, 2, or 3 stones.\n");
            continue;
        }
```

```
        if (playerMoveChoice > stones) {
            printf("❌ Not enough stones left!\n");
            continue;
        }

        stones -= playerMoveChoice;

        if (stones == 0) {
            printf("🎉 You removed the last stone. You win!\n");
            return;
        }

        printf("You removed %d stones. %d remaining.\n", playerMoveChoice, stones);
        printf("AI is thinking...\n");
        sleep(1);  // optional pause for realism

        // AI's turn
        int aiMove = bestMove(stones);
        stones -= aiMove;
        printf("🤖 AI removed %d stones.\n", aiMove);

        if (stones == 0) {
            printf("💀 AI removed the last stone. You lose!\n");
            return;
        }
    }
}
```

APPENDIX B – screenshot

```
🪨 Welcome to the 25-Stones Nim Game!
Rules:
 - There are 25 stones.
 - You and the AI take turns.
 - On each turn, remove 1, 2, or 3 stones.
 - The player who removes the last stone
     wins!


-----------------------------------
Stones remaining: 25
Your turn! Enter stones to remove (1-3): 2
You removed 2 stones. 23 remaining.
AI is thinking...
🤖 AI removed 3 stones.

-----------------------------------
Stones remaining: 20
Your turn! Enter stones to remove (1-3): 1
You removed 1 stones. 19 remaining.
AI is thinking...
🤖 AI removed 3 stones.

-----------------------------------
```

```
Stones remaining: 16
Your turn! Enter stones to remove (1-3): 3
You removed 3 stones. 13 remaining.
AI is thinking...
🤖 AI removed 1 stones.

----------------------------------
Stones remaining: 12
Your turn! Enter stones to remove (1-3): 1
You removed 1 stones. 11 remaining.
AI is thinking...
🤖 AI removed 3 stones.

----------------------------------
Stones remaining: 8
Your turn! Enter stones to remove (1-3): 2
You removed 2 stones. 6 remaining.
AI is thinking...
🤖 AI removed 2 stones.

----------------------------------
Stones remaining: 4
Your turn! Enter stones to remove (1-3): 1
You removed 1 stones. 3 remaining.
AI is thinking...
🤖 AI removed 3 stones.
💀 AI removed the last stone. You lose!
```

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